

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology, Kanpur	7.9/10.0
2022	XII(CBSE)	Rukmani Birla Modern High School	86.2%
2020	X(CBSE)	Rukmani Birla Modern High School	87.2%

Scholastic Achievements

- Secured All India Rank **4042** in **Joint Entrance Examination - Advanced 2023** among **250,000** selected candidates
- Secured All India Rank **11977** in **Joint Entrance Examination - Mains 2023** among **1.1 million** listed candidates
- Secured an All India Rank of **104** in the **Undergraduate Entrance Examination - 2023** among **10,000** candidates
- Received **Merit Certificate from CBSE** for being in the **top 0.1% scorers of Mathematics-Standard** in 10th Board

Key Projects

Frequency-Based Image and Audio Processing | Course Project (EE200) | Prof. Tushar Sandhan

(Jun'25-Jul'25)

Objective	● To develop a signal/image processing pipeline for hybrid image fusion and audio denoising using Python
Approach	● Applied 2D DFT with custom tunable filters to effectively isolate key spatial frequency components in images ● Used Welch's method and band-stop filters (1–10 kHz) to suppress noise while preserving audio integrity ● Used Matplotlib and scipy.signal for waveform analysis, PSD estimation, and frequency-domain diagnostics
Results	● Achieved over 90% frequency isolation accuracy, producing fused images with distinct clarity at varying distance ● Successfully isolated the noise of an unwanted instrument from a song, using Bode plots to confirm signal fidelity

Stock Price Prediction using LSTM | Self Project | 

(May'25-Jun'25)

Objective	● To build a RNN-based model to predict Google stock prices using a recent 5-year window for higher relevance.
Approach	● Preprocessed 5,017 rows × 6 features of time series stock data using MinMaxScaler for LSTM readiness ● Built a deep LSTM architecture with 4 stacked layers × 100 units each and 20% dropout to reduce overfitting ● Trained the model over 200 epochs with batch size 64 , incorporating validation and model checkpointing
Results	● Forecasted GOOG stock movements across 50+ future trading days, aligned visually with actual market behavior ● Obtained R ² scores of 0.997 (train) , 0.914 (val) , and 0.822 (test) , indicating strong model generalization ● Achieved RMSE of 1.61 (train), 1.75 (val), and 1.79 (test), showing strong precision on actual price scale units

Devanagari Optical Character Recognition | Self Project | 

(May'25-Jun'25)

Objective	● To develop a high-accuracy OCR system for handwritten Devanagari characters using CNN-based learning
Approach	● Preprocessed 32×32 grayscale images with label one-hot encoding, employing ReLU and softmax activations ● Applied early stopping and validation monitoring to optimize training performance and generalization ● Built a 7-layer CNN (3 Conv, 2 MaxPooling, 1 Dense, 1 Output) trained with Adam optimizer for 20+ epochs
Results	● Achieved 98.3% accuracy in classifying 46 different Devanagari characters with real-time data augmentation

Machine Learning & Neural Networks Applications | Coursera | 

(May'25-Jun'25)

Approach and Results	● Compressed images efficiently using K-Means Clustering by reducing color palette from 1000+ to 16 colors ● Used decision trees to classify poisonous mushrooms; implemented collaborative filtering for recommendations ● Applied tree ensembles & XGBoost for Heart Failure Prediction , achieving accuracy rates of 88% & 86% ● Trained a deep model to land a lunar lander safely on a moon landing pad using a reinforcement model
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Technical Skills

- **Programming Languages:** C | C++ | Python | $\LaTeX$ | MATLAB | MySQL
- **Software:** Git | Github | $\LaTeX$
- **Libraries:** Numpy | Pandas | Matplotlib | Keras | SKLearn | TensorFlow | PyTorch | OpenCV | PIL

Position of Responsibility

Coordinator | Quiz Club | IIT Kanpur

(Apr'25-Present)

Leadership	● Managing a budget of over 80K INR and allocating funds strategically among collaborations and logistics
Initiative	● Conducted the Diversity Quiz with 200+ participants nationwide, marking a 33% rise from last year ● Invited an external quizmaster to host a campus-wide quiz with over 100 participants attending
Impact	● Drove 200%+ growth in student participation by revamping event formats and outreach channels

Relevant Courses

†: Online Course

Economy, Society and Public Policy	Fundamentals of Computing	Linear Algebra
Supervised Machine Learning †	Advanced Learning Algorithms †	Unsupervised Machine Learning †

Extra-Curriculars

Cultural	● Part of the Quiz Cup Contingent at the Inter IIT Cult Meet 6.0 that came 5th overall out of 24 IITs ● Part of the Quiz Cup Contingent in the Inter IIT Cult Meet 7.0 that came 6th overall out of 24 IITs ● Represented IIT Kanpur in the Inter-IIT-IIM Quiz Fest Nihilanth 2025 among various IITs and IIMs
Social Work	● Volunteered in Ritambhara, IITK's Fashion Carnival, as the primary liaison for invited judges and guests ● Active blood donor , regularly contributing to community's well-being through voluntary blood donations